SA Normality Report

Naspers Limited (NPN.JO)

02 Jan 2018 to 28 Oct 2026 · simple returns · 2 205 observations · ZAR

What this means

This analysis covers Naspers Limited (NPN.JO) daily returns. Normality is rejected at the 5% level (Jarque-Bera p < 0.0001). Returns are positively skewed and fat-tailed (skew 0.38, excess kurtosis 8.49), so large moves are more frequent than a normal model implies. At the 99% level, historical VaR is 6.47% versus 5.83% under a normal fit, so the normal model understates the daily tail loss by 1.1x. 13 days fell more than three standard deviations below the mean, versus about 3.0 expected under normality. The worst day was -17.70% and maximum drawdown was -62.8%. A Student-t fit with 3.9 degrees of freedom beats the normal fit by AIC 387. This describes the selected historical sample, not a forecast.

Diagnostics

OBSERVATIONS	SKEWNESS	EXCESS KURTOSIS
2 205	0,38	8,49

ANNUALISED VOL	TOTAL RETURN	JARQUE-BERA P
40.1%	54.5%	< 0.0001

Tail risk: historical vs normal

Measure	Historical	Normal	Hist. / normal
95% VaR	3.74%	4.10%	0.9x
99% VaR	6.47%	5.83%	1.1x
95% CVaR	5.43%	5.16%	1.1x
99% CVaR	8.95%	6.68%	1.3x

13 days fell more than three standard deviations below the mean, versus about 3 expected under normality. Worst day: -17.70%. Maximum drawdown: -62.8%.

Student-t fit: nu = 3.9, maximum likelihood, delta AIC 387 vs normal.

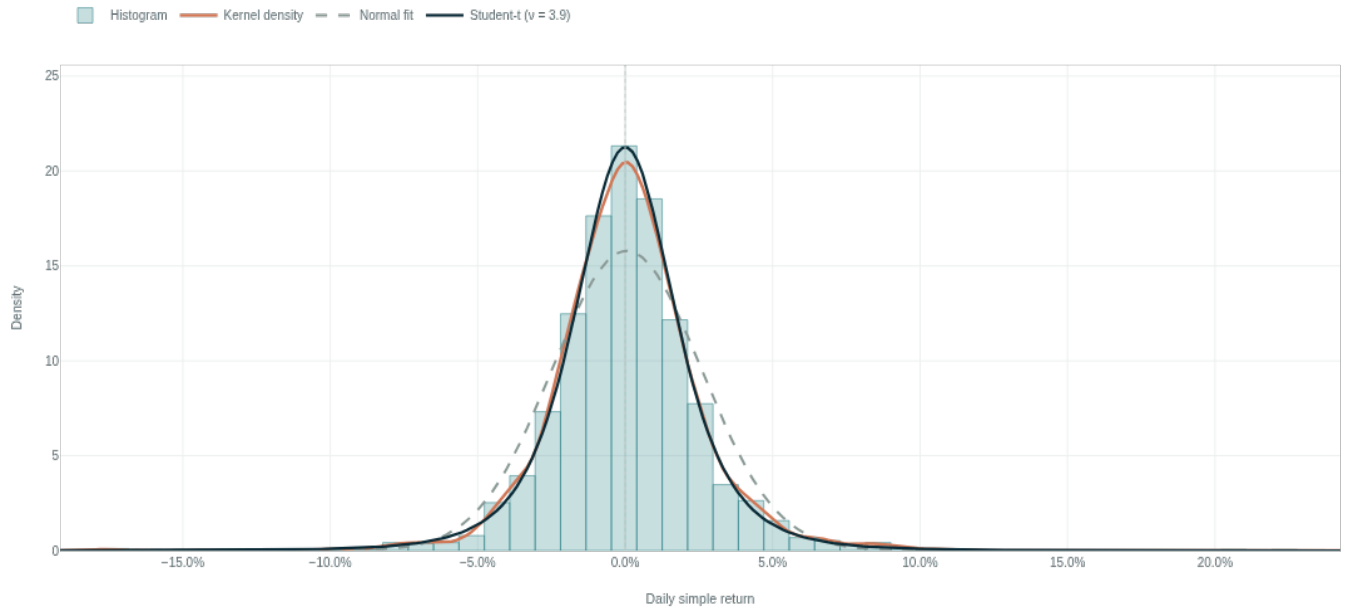
Price history

Adjusted close in ZAR



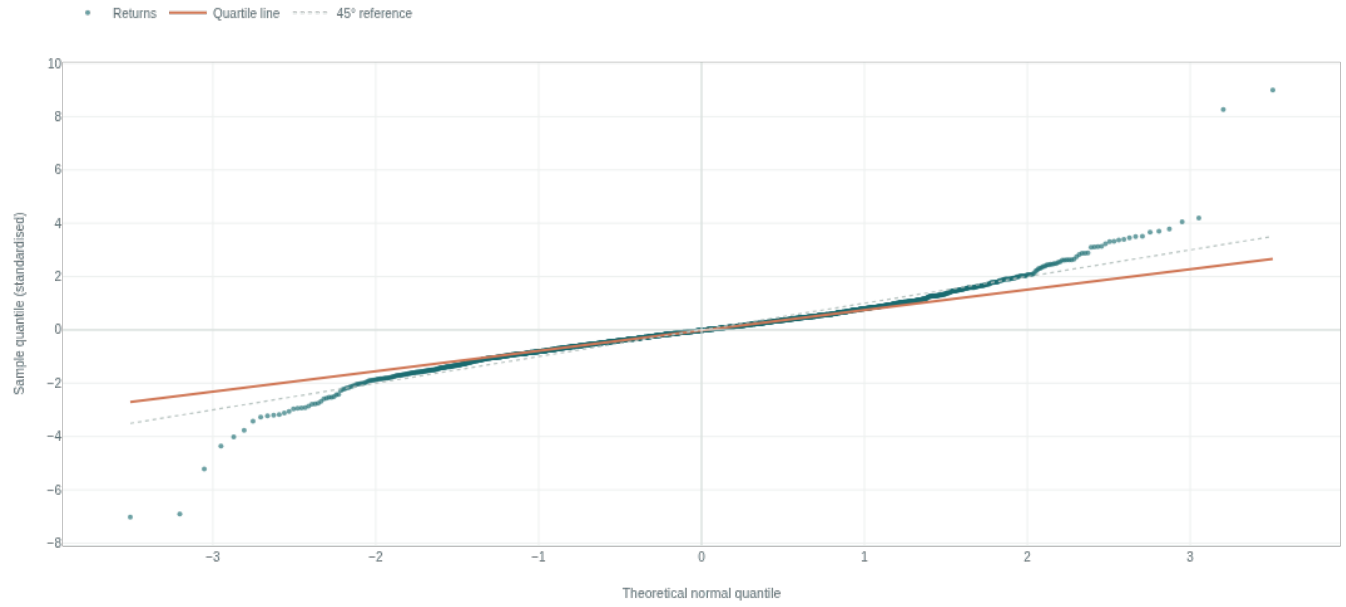
Distribution of returns

Histogram, kernel density, normal fit and Student-t fit



Normal Q-Q plot

Standardised returns against theoretical normal quantiles



Methodology

VaR and CVaR are 1-day historical estimates from the selected sample. Normal estimates use the sample mean and standard deviation. The Student-t fit is estimated by maximum likelihood, with a moment approximation if optimisation fails. Returns use adjusted close; JSE prices are converted from cents to rand. Data from Yahoo Finance (delayed). This is a historical description, not a forecast or investment advice.